

Week 6-Semester 2

Bond Market

Readings: Burton, M, Nesiba, R., and Brown, B. 2015. An introduction to Financial markets and Institutions. Second Edition. Routledge, Taylor & Francis Group. London & New York.

- In this lecture we will discuss the following:
 - Why do we need the bond market?
 - Governmental bonds
 - Corporate bonds
 - Bonds terminology
 - Key factors that drive bond prices
 - Bond valuation

1- Why do we need the bond market?

- Bonds are used to finance governments
- Government bond yields act as reference points and keep governments accountable
- Bonds also finance corporations

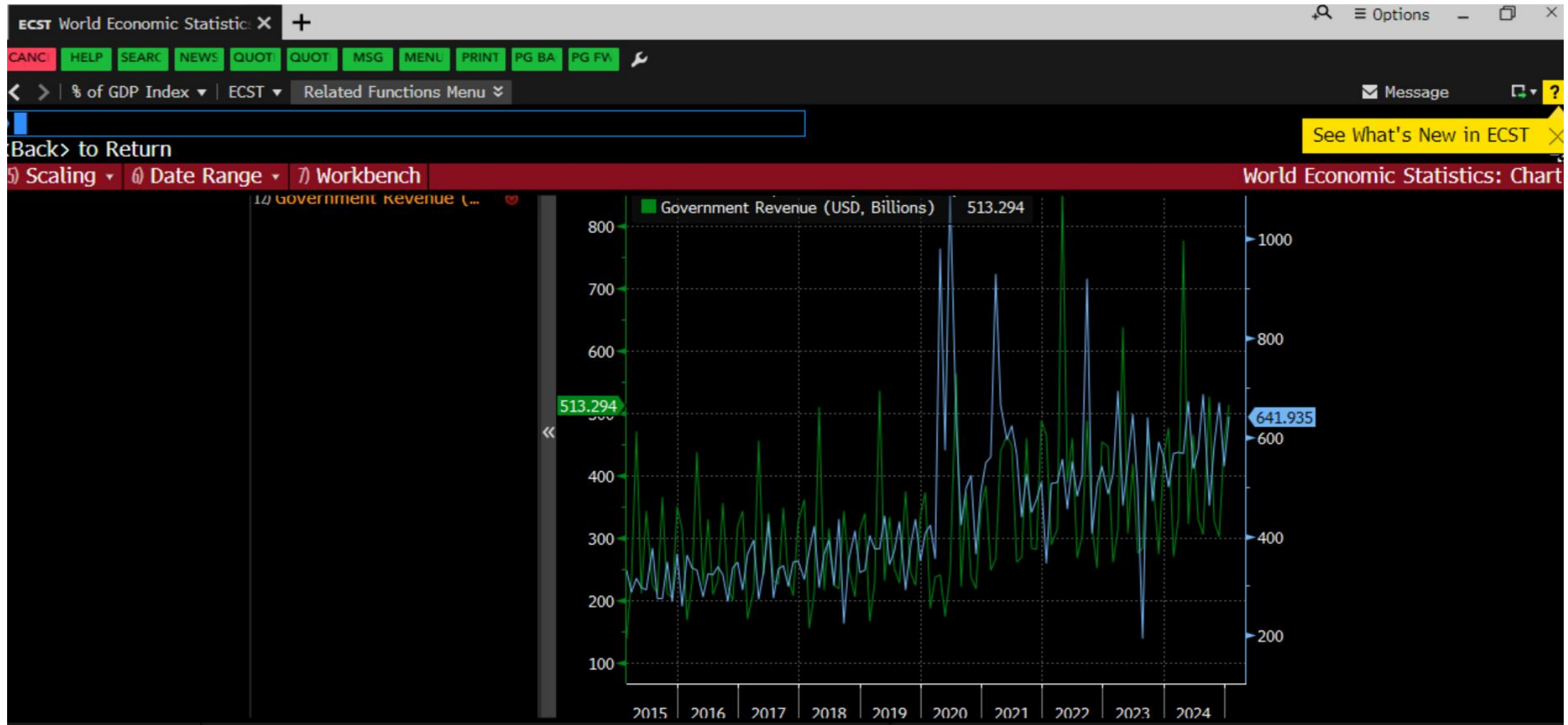
A- Governmental Bonds (gilts)

- Treasury bonds in the US
- They carry virtually no default risk since the Treasury can print money to payoff the debt.
- Very low interest rates, often considered the risk-free rate (although inflation risk is still present)
- Gilts in the UK
 - the original certificates issued by the British government had gilded edges

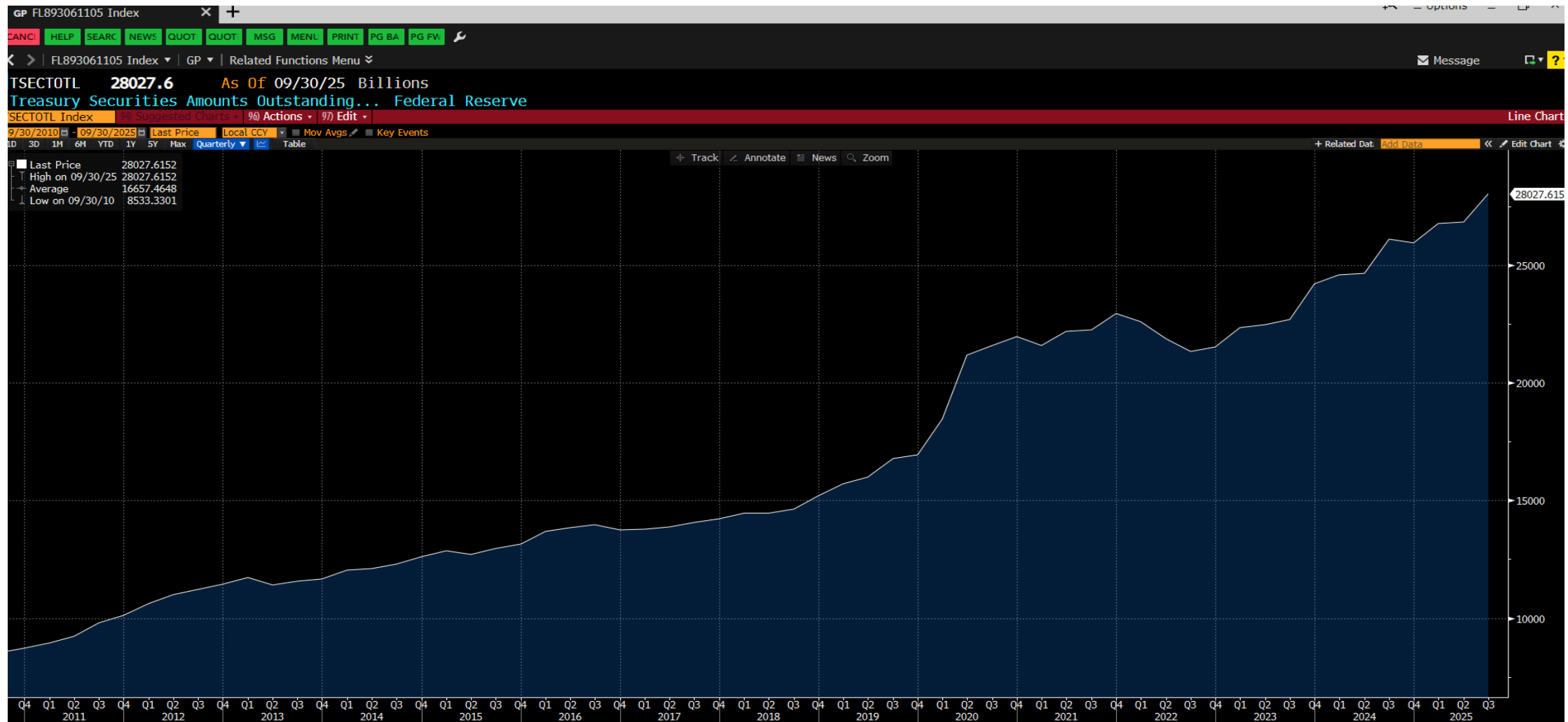
Mainly two types:

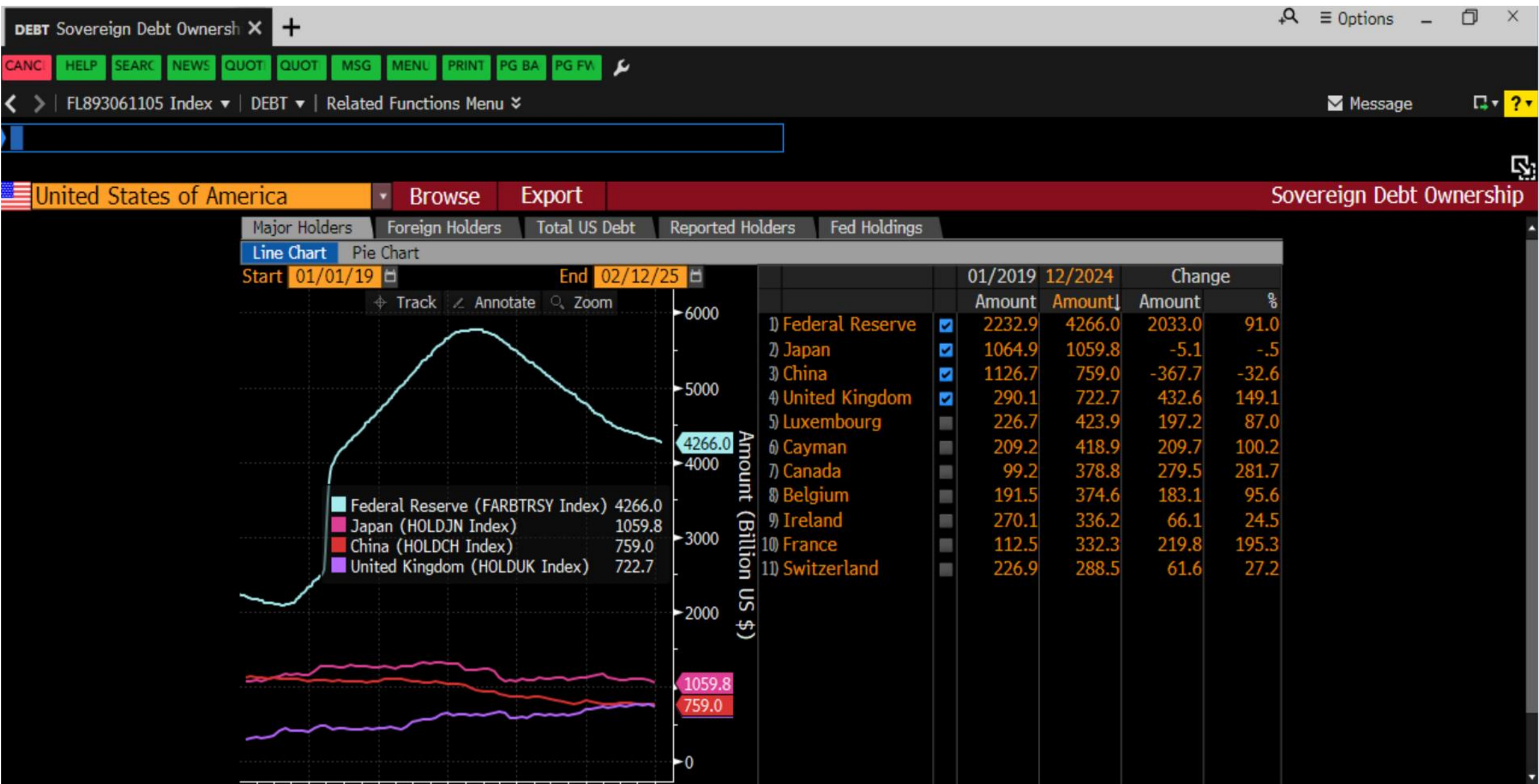
- Conventional, issued in nominal terms
- indexed to inflation
 - called Treasury Inflation-Protected Securities (TIPS) in the US, index-linked gilts in the UK.

US Federal Budget deficit/surplus



Debt outstanding-US





B-Corporate Bonds

- Very similar to Treasury bonds with its components
- Pay interest semi-annually
- Degree of risk varies with each bond, even from the same issuer.
 - the interest rate varies with level of risk.
- The degree of risk ranges from low-risk (AAA) to higher risk (BBB). Any bonds rated below BBB are considered sub-investment grade debt.
- Even though the bonds of a corporation are less risky than its equity, investors still have risk: price risk and interest rate risk

- Financial Guarantees for Corporate Bonds
- Some debt issuers purchase financial guarantees to lower the risk of their debt.
- The guarantee provides for timely payment of interest and principal and are usually backed by large insurance companies.
- As it turns out, not all guarantees actually make sense!
 - In 1995, JPMorgan created the credit default swap (CDS), a type of insurance on bonds.
 - In 2000, regulatory oversights on CDSs were eliminated.
 - By 2008, the CDS market was over \$62 trillion!
 - 2008 losses on mortgages lead to huge payouts on this insurance.

C-Bond Terminology

Term	Definition
Annual Interest payment	a nominal sum to be paid each year in two half-yearly installments until the bond is redeemed.
Coupon interest rate	% of the face value paid annually
Current yield. (i_c) Or Running yield	The coupon interest payment divided by the current market price of the bond (C/P)
Face value Or Par value	The holder of the bond will receive the face value from the issuer when the bond matures
Maturity	The number of years or periods until the bond matures and the holder is paid the face amount.
Yield to maturity Or redemption yield	The yield an investor will earn if the bond is purchased at the current market price and held until maturity.

How does an investor make money on bonds?

An investor who holds £1,000 nominal of 1.5% Treasury Gilt 2035 will receive two coupon payments of £7.50 each on 22 January and 22 July, totalling to £15 each year.

2035 is the year of redemption (10 year bond)

1.5% is the coupon rate, which is annual ($1.5\% = 0.015$)

$£1,000 \times 0.015 = £15$ is the coupon payment every year = C

How much would you pay today to buy this bond?

Is it more or less than £1,000?

It depends on current interest rates, time to maturity, inflation, inflation expectations...

Based on these you'd have a "required rate of return" in mind and that would determine the price you'd be willing to pay

Bond Yield vs. Price

As bond prices increase, bond yields fall . Why?

Suppose an investor purchases a bond that matures in five years with a 10% annual coupon rate and a face value of £1,000.

- If interest rates rise above 10%
 - The original bond still only makes a coupon payment of £100, which would be unattractive to investors who can buy bonds that pay more than £100.
 - If the original bond owner wants to sell the bond, the price must be lowered so that the coupon payments and maturity value will bring a higher return.
- If interest rates were to fall , the bond's price would rise because its coupon payment is more attractive.
 - For example, if interest rates fell to 7.5% for similar investments, the bond seller could sell the bond for £1,101.15. The further rates fall, the higher the bond's price will rise, and the same is true in reverse when interest rates rise.

Current Yield Calculation

Suppose a bond is selling at £921 = price = P. What is the current yield for this bond with a face value of £1,000, and a coupon rate of 1.5%?

$$C \text{ (coupon)} = £15$$

Answer:

$$i_c = C / P$$

$$i_c = £15 / £921 = 0.0163 = 1.63\%$$

Another example:

What is current (or running) yield on a bond trading in the secondary market at £120 which pays a coupon of £6?

$$i_c = C / P = 6/120 = 0.05 = 5\%$$

Yield to maturity calculation

Current yield is different from redemption yield (Yield to maturity) because it only considers a single year return

- Yield to maturity takes account of any capital gain/loss at redemption.

e.g. a bond with 10 years to redemption with a coupon of £6 is currently trading at £120 (i.e. price)

The current yield is $6/120 = 0.05 = 5\%$.

The holder receives the face value of £100 at redemption, leaving her with a capital loss of £20. In percentage terms, capital loss = $£20/£120 = 0.167 = 16.7\%$.

If 10 years to maturity, the annualized percentage capital loss = $0.167/10 = 0.0167$

Yield to maturity = $0.05 - 0.0167 = 0.033 = 3.3\%$

- Alternatively, simply sum the coupon payments as $£6 \times 10 = £60$, subtract the capital loss of £20 = £40, divide by time to redemption = $40/10 = 4$ and divide by 120 = 0.33 or 3.3% .

Yield to maturity calculation

In these calculations, we disregarded the present value vs future value distinctions.

How can we calculate the value of a 2-year bond with face value of £1,000 and 5% coupon rate. You need to know the market interest rate to decide how much it is worth today.

This becomes the basis for calculating the bond prices

Note: the interest rate in the formula is NOT the coupon rate, but the prevailing market rate.

Remember:

PV = today's (present) value

CF = future cash flow (payment)

i = the interest rate

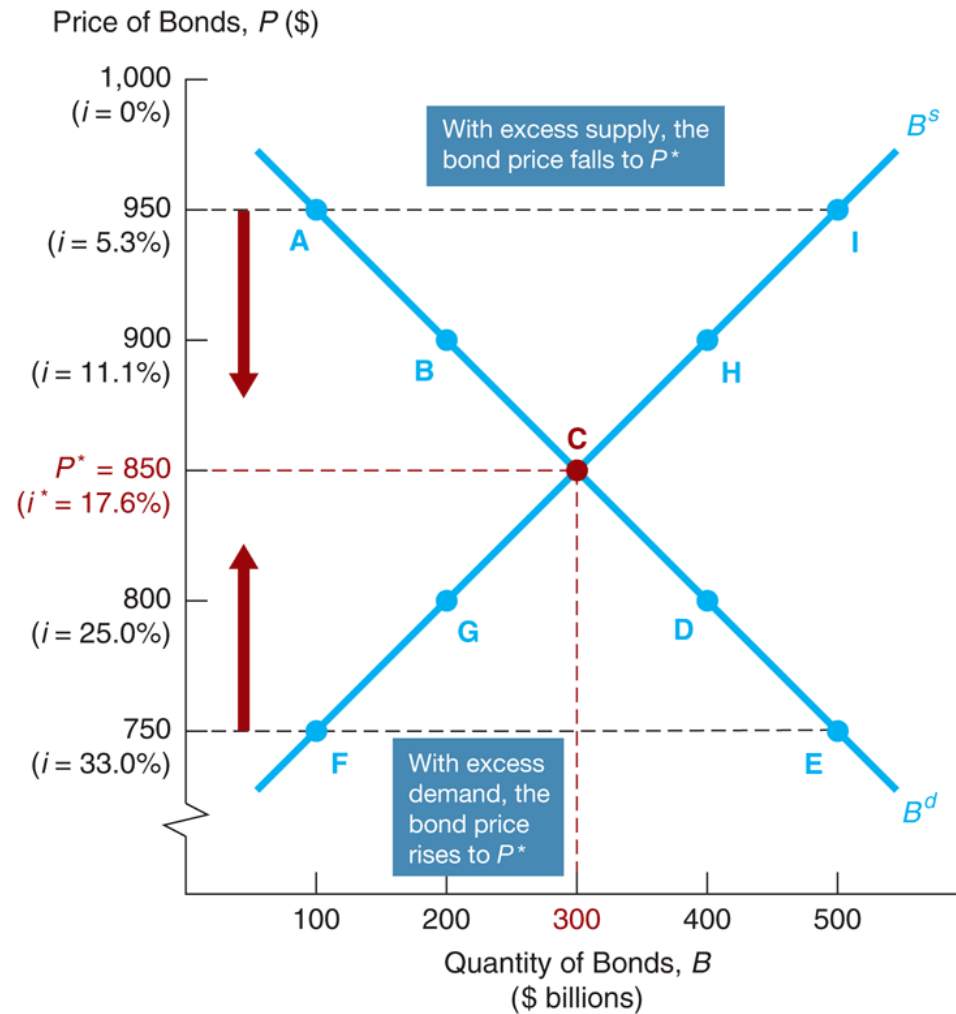
$$PV = \frac{CF}{(1 + i)^n}$$

Yields and maturities

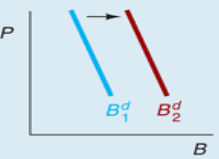
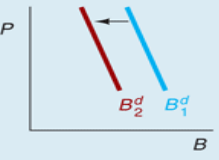
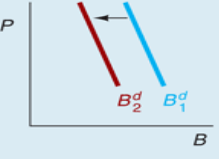
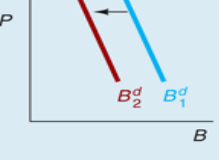
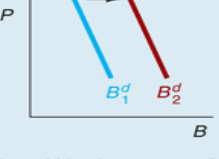
- Bonds are classified into types according to their initial maturities.
- Shorts – having up to 5 years to redemption.
- Mediums – having up to 15 years to redemption.
- Longs – having more than 15 years to redemption.
- Ultra longs – longs with at least 40 years to redemption.
- Because bonds pay a nominal coupon, the real value of which is reduced by inflation, investors will demand higher yields from bonds with a long time to maturity, reflecting the greater uncertainty about future rates of inflation.
- <https://www.bloomberg.com/markets/rates-bonds/government-bonds/uk>

2- Bond valuation

- The main 3 factors that drive bond yields are: the creditworthiness of the borrower, inflation rate and short-term interest rates.
- The bond market instils discipline in governments as creditworthiness affects borrowing costs for them.
- Short-term borrowing tends to be cheaper but could be riskier for borrowers than long-term borrowing, as it relies on the sentiment of lenders.
- Given that repayments to bondholders are fixed, inflation can corrode the purchasing power of these fixed payments, increasing yields.
- Yields on US government bonds can affect other bonds.



Factors That Shift the Demand Curve for Bonds

Factors That Shift the Demand Curve for Bonds			
Variable	Change in Variable	Change in Quantity Demanded at Each Bond Price	Shift in Demand Curve
Wealth	↑	↑	
Expected interest rate	↑	↓	
Expected inflation	↑	↓	
Riskiness of bonds relative to other assets	↑	↓	
Liquidity of bonds relative to other assets	↑	↑	
<i>Note:</i> Only increases in the variables are shown. The effect of decreases in the variables on the change in demand would be the opposite of those indicated in the remaining columns.			

UK Bond Demand Hits Record as Easing Bets Outweigh Supply Fears

- UK inflation-linked sale gets more than £56 billion of orders
- Macro picture is supportive of gilts, says Toronto Dominion

By James Hirai

UK Debt Finds Attraction

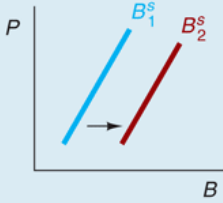
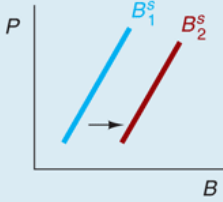
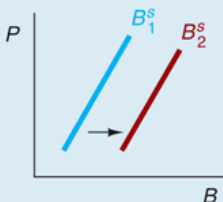
Investors place record orders above £56 billion for UK inflation-linked debt



Source: Bloomberg

Bloomberg

Factors That Shift the Supply of Bonds

Factors That Shift the Supply of Bonds			
Variable	Change in Variable	Change in Quantity Supplied at Each Bond Price	Shift in Supply Curve
Profitability of investments	↑	↑	
Expected inflation	↑	↑	
Government deficit	↑	↑	

Note: Only increases in the variables are shown. The effect of decreases in the variables on the change in supply would be the opposite of those indicated in the remaining columns.

WB World Bond Markets

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RegionalExportSettings

91) Bonds92) Spreads93) Curves

Region		2 Year		3 Year		5 Year		7 Year		10 Year		15 Year		30 Year	
1) Americas															
10) United Stat...		4.295	-1.0	4.317	-1.1	4.397	-0.5	4.483	+0.8	4.562	+1.2			4.790	+2.4
11) Canada		2.798	+0.2	2.774	+0.1	2.891	+0.7	3.019	+1.0	3.202	+1.5			3.380	+1.8
12) Brazil (USD)		5.093	+1.2	5.431	+5.0	5.859	+0.9	6.306	+1.6	6.607	+1.6	6.819	-0.1	7.473	+2.1
13) Mexico (US...		5.017	+3.6	5.283	+2.2	5.702	+1.6	6.153	+1.2	6.454	+1.3			7.215	+1.4
2) EMEA															
19) United King...		4.262	+3.1	4.047	+3.1	4.292	+4.4	4.417	+4.7	4.605	+5.0	4.907	+4.8	5.187	+4.4
20) France		2.317	+3.5	2.466	+4.3	2.744	+5.5	2.972	+5.5	3.258	+5.5	3.513	+5.3	3.781	+4.8
21) Germany		2.163	+4.0	2.174	+4.0	2.309	+4.6	2.359	+4.9	2.537	+4.6	2.768	+4.2	2.781	+3.8
22) Italy		2.435	+4.8	2.578	+5.8	2.927	+6.8	3.256	+6.8	3.613	+6.8	3.959	+6.9	4.297	+6.6
23) Spain		2.370	+5.6	2.487	+5.6	2.684	+5.7	2.838	+5.6	3.153	+5.8	3.495	+5.7	3.838	+5.2
24) Portugal		2.085	+3.3	2.133	+5.4	2.371	+5.2	2.623	+5.8	2.962	+5.7	3.265	+5.7	3.658	+6.3
25) Sweden		1.984	+3.2			2.127	+3.4	2.195	+3.2	2.380	+2.9	2.510	+2.7		
26) Netherlands		2.179	+4.0	2.222	+4.1	2.386	+4.4	2.520	+4.8	2.721	+4.9	2.882	+4.0	2.891	+3.9
27) Switzerland		0.187	+2.9	0.249	+2.0	0.322	+1.8	0.404	+1.3	0.481	+1.3	0.569	+1.8	0.584	+2.1
28) Greece		2.229	+5.5	2.424	+5.2	2.724	+5.7	3.032	+5.0	3.371	+6.4	3.617	+6.2	3.985	+6.4
3) Asia/Pacific															
29) Japan		0.809	-0.3	0.859	-0.1	1.066	-0.3	1.195	--	1.421	+0.1	1.980	+0.4	2.321	-1.1
30) Australia		3.913	+0.7	3.937	+0.9	4.099	+0.8	4.347	+1.6	4.522	+1.9	4.716	+1.7	5.008	--
31) New Zealand		3.664	+2.7	3.832	+2.0	4.124	+3.0	4.398	+3.4	4.569	+3.7	4.833	+3.7	5.196	+3.7

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VIX Index

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Line Chart

03/31/1989 - 08/14/2014

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Security/Study

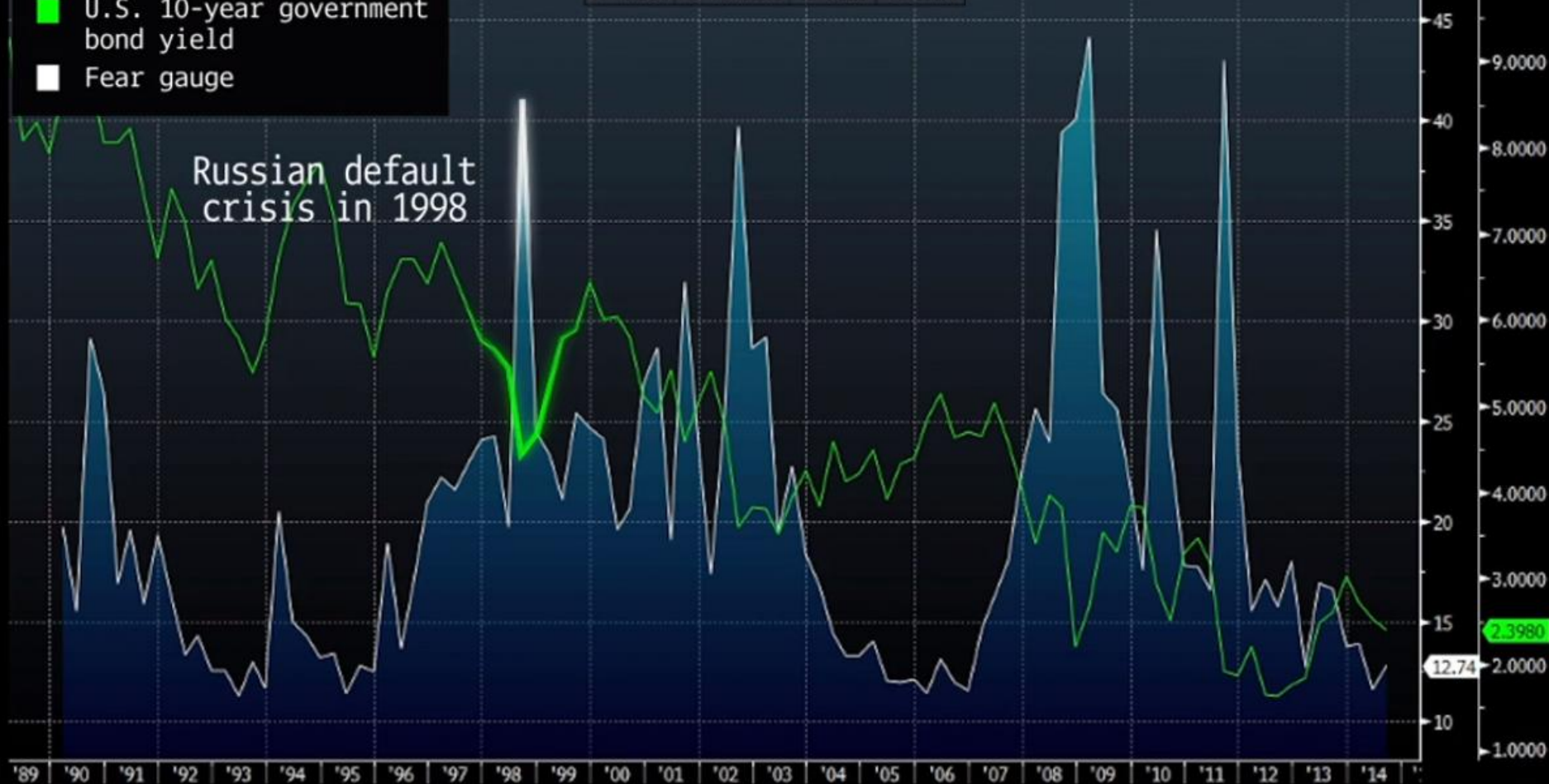
Event

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- U.S. 10-year government bond yield
- Fear gauge

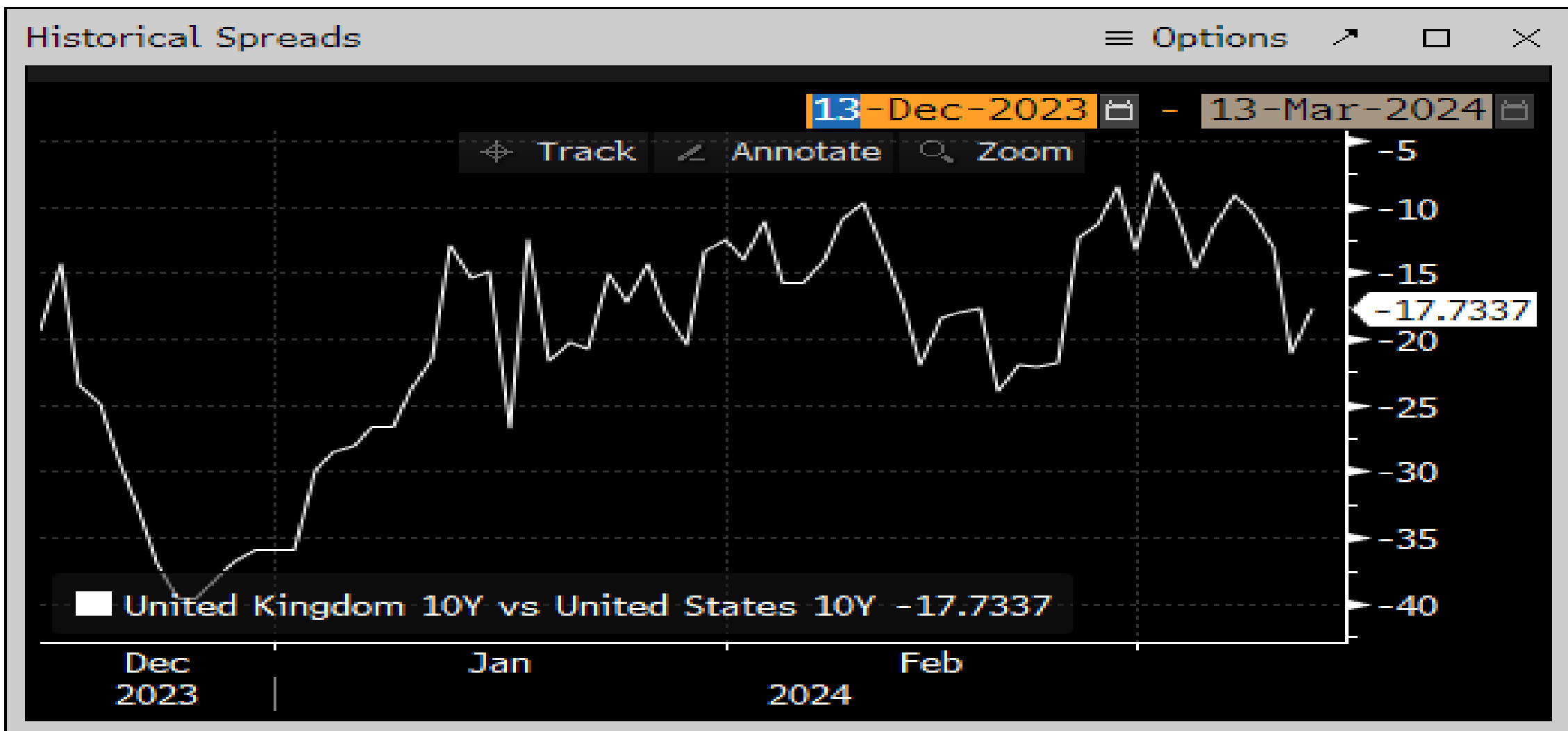
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Russian default crisis in 1998



2.3980

12.74



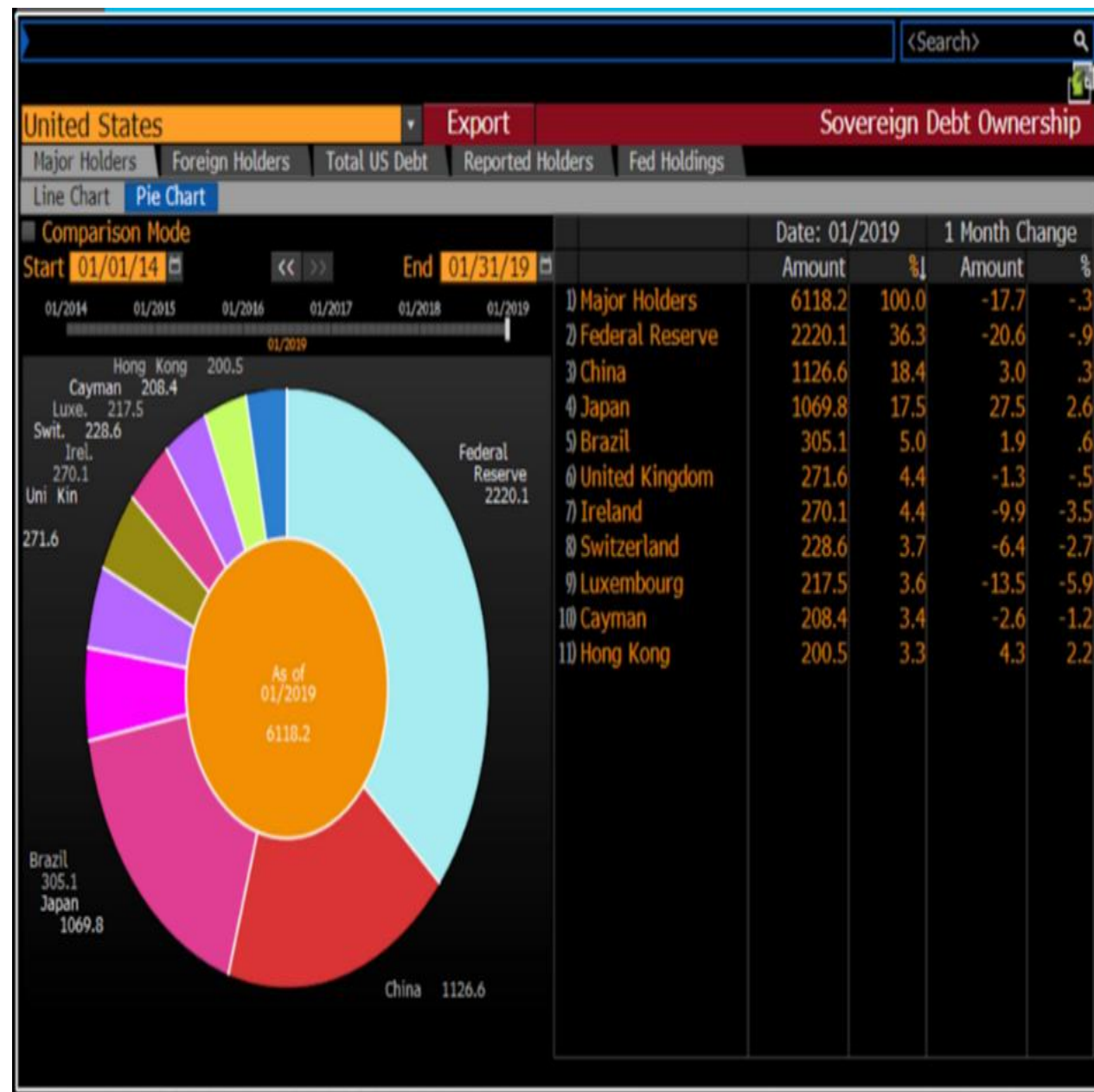
Source: Bloomberg.

Conclusion

- Fixed income is another word for the bond market.
- The term fixed income is derived from the fixed nature of the bond fixed payments.
- The growth of governments deficit has been the main driving factor for the growth of the \$130T world bond market (end of 2025)
- The US treasury bond is perceived to be the safest , most liquid security.
- Yields are used to compare different bonds.
- The three biggest factors deriving bond yields are the creditworthiness of the borrower, inflation, and short-term interest rates.
- As repayments to bondholders are fixed, inflation will have a negative impact on the fixed bond repayments and hence raising yields.

QI: At the end of January 2019, the total traded US debt is \$16T. According to the table, approximately what percent of US debt does China own at this time?

- a- 17%
- b-18.4%
- c-7%
- d-3%



Q2: Which one of the following actors benefits when interest rates go up?

- a- An investor who is about to buy bonds
- b- An investor who already owns bonds
- c- A company with a fixed-rate loan
- d- A company about to secure a fixed-rate loan

Q3: When investors doubt the creditworthiness of a borrower, what should happen to the price and yield of the bond?

a-Price goes down, yields go down.

b- Price goes up, yields go up

c- Price goes up, yields go down

d-Price goes down, yields go up

Q4: A drop in which of the following measures would typically send a government bond price down?

a- Government borrowing

b- creditworthiness

c- Inflation

d-Interest rates

Q5: An investor is comparing two bonds of similar structure from the same issuer. Which bond should the investor buy?

- a- The bond with the highest coupon and the lowest yield.
- b-The bond maturing last as that has the lowest risk
- c-The bond with the highest yield if the two bonds have the same maturity date.
- d-The bond with the highest price as that offers the highest return